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Multispectral Remote Sensing Image Unmixing Method Based on Unsupervised Optimization-Enhanced CNN

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***Abstract*—Multispectral remote sensing images typically contain a large number of mixed pixels, which poses significant challenges for image interpretation. However, existing deep learning-based unmixing methods heavily rely on the completeness of the training data. This paper proposes an Unsupervised Optimization-enhanced Convolutional Neural Network (UO-CNN) method that integrates the unsupervised prior information from Fuzzy C-Means clustering with the fine modeling capability of an enhanced CNN, thereby achieving a synergistic combination of high accuracy and strong generalization performance. Experiments demonstrate that compared to traditional CNN methods, UO-CNN improves the unmixing correlation coefficient R by 3.4% and reduces RMSE by 25% in standard scenarios; in generalization scenarios containing unknown land cover categories, it still demonstrates good performance.**

Keywords—spectral unmixing, Fuzzy C-Means, unsupervised learning, enhanced CNN

I. INTRODUCTION

The spatial resolution of remote sensing images is constrained by the physical characteristics of sensors, where individual pixels often contain mixed spectral information from multiple land cover types, forming mixed pixels. This phenomenon severely restricts the accuracy of fine land cover classification. Subpixel mapping (SPM) technology obtains abundance information of various land cover categories through spectral unmixing and reconstructs high-resolution land cover distribution maps[1]. The effectiveness of unmixing methods directly affects the performance of subpixel mapping.

Traditionally, spectral unmixing methods fall into three categories: geometry-based (such as VCA and N-FINDR), statistics-based (such as ICA and NMF), and physics-based (such as FCLS). Although theoretically well-founded, these methods commonly suffer from overreliance on linear mixing assumptions, dependence on pure endmember priors, and limited spatial information utilization.

In recent years, the CNN-based spectral unmixing methods have effectively overcome many limitations of traditional approaches through end-to-end learning that automatically extracts deep features[5]. However, existing deep learning unmixing methods face a critical challenge: strong dependence on the completeness of land cover categories in the training data. When test scenarios contain land cover types that do not appear in the training set, the model performance degrades dramatically[7].

To address this limitation, this paper proposes a UO-CNN unmixing method that organically combines the generalization capability of FCM with the fine modeling capability of CNNs. Through globally optimized FCM preprocessing, this method provides reliable initial estimates for unknown endmember categories, and subsequently employs a CNN to learn the complex mapping relationship from FCM results to true abundance. Furthermore, experimental results indicate that this method can effectively improve unmixing accuracy while demonstrating strong generalization capability.

II. UO-CNN METHOD

As shown in Fig. 1, the core architecture of the UO-CNN proposed in this paper consists of three components.

The first component is the FCM preprocessing module. The FCM clustering results can be optimally matched with the reference abundance data, and cluster center information conforming to the actual ground truth distribution can be stably extracted, thereby providing essential prior knowledge for the subsequent network.

The second component is the FCM-SVM alignment module. Using high-resolution abundance estimates generated by the SVM as the reference benchmark, a correlation matrix is constructed through the correlation coefficients and mean squared error metrics.

The third component is the enhanced dual-input deep convolutional network, which can simultaneously process the raw multispectral data and the aligned FCM abundance prior information. This network employs a physically inspired

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endmember-abundance dual-branch architecture with multi-scale feature extraction and residual learning. Additionally, this structure enforces physical abundance constraints through Softmax activation.

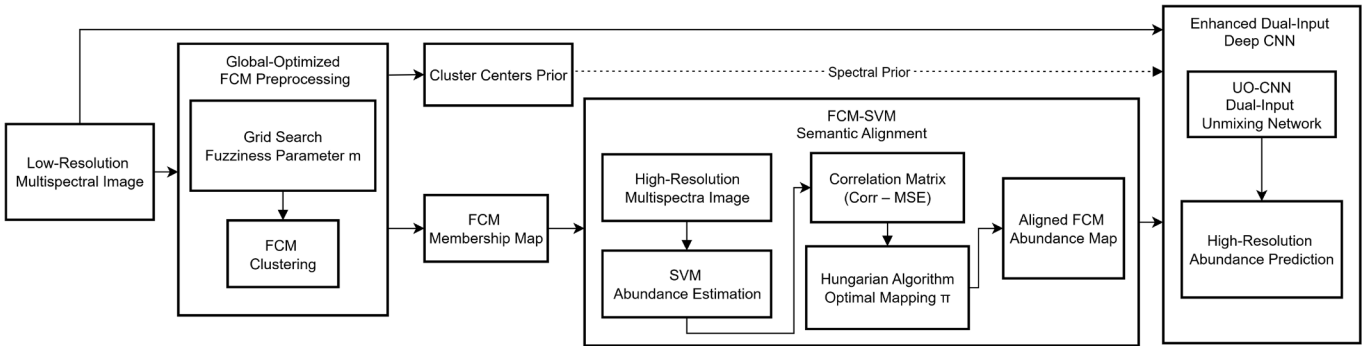


Fig. 1. Flowchart of the proposed unsupervised object-based Convolutional Neural Network

A. Unsupervised preprocessing based on FCM

UO-CNN employs an adaptive gradient-based optimization strategy to dynamically determine the optimal fuzziness parameter m during training[2]. For images with B bands, the FCM objective function is defined as[3]:

$$\min_{F} \sum_{n=1}^{N} \sum_{k=1}^{C} \left(\mu_k(n)\right)^{m} \| y_n - v_k \|^2 \qquad (1)$$

subject to: $\mu_k(n) \in [0,1]$, $\sum_{k=1}^{C} \mu_k(n) = 1$.

Adaptive Optimization Algorithm for m:First,Set $m^{(0)} = 2.0$, learning rate $\eta = 0.01$, tolerance $\epsilon = 10^{-5}$. Then,at iteration t, compute gradient calculation:

$$\frac{\partial F}{\partial m^{(t)}} = \sum_{n=1}^{N} \sum_{k=1}^{C} \left(\mu_k(n)\right)^{m^{(t)}} \ln\left(\mu_k(n)\right) \| y_n - v_k \|^2 \qquad (2)$$

$$m^{(t+1)} = m^{(t)} - \eta \cdot \frac{\partial F}{\partial m^{(t)}} \qquad (3)$$

Terminate when $\left|m^{(t+1)} - m^{(t)}\right| < \epsilon$.This strategy ensures m converges to a task-specific optimum, enhancing reproducibility by providing explicit update rules and hyperparameters.

B. FCM-SVM alignment mechanism

The alignment module establishes correspondence between FCM clustering results and high-resolution reference abundance data. To address concerns about using SVM-generated abundances as ground truth, we implement a multi-resolution validation framework that quantifies and minimizes potential biases[4].

The high-resolution reference abundances are generated through a dual-method validation approach:Primary SVM classification is performed on high-resolution imagery (10m Sentinel-2), producing initial abundance estimates. A secondary verification using Maximum Likelihood Classification (MLC) and Random Forest (RF) is conducted on the same high-resolution data. Only pixels with consistent classification results (agreement > 95%) between both methods are aggregated to create low-resolution reference abundances.

The aggregation process from high to low resolution follows:

$$REF_{low(i,j)} = \left(\frac{1}{n^2}\right) \sum REF_{high(ni+p,nj+q)} \qquad (4)$$

This dual-validation at high resolution significantly reduces systematic errors before aggregation, ensuring that the reference abundances used for FCM alignment are more reliable than single-method approaches. The Hungarian algorithm then determines the optimal assignment $\pi: \{1,\ldots,C\} \rightarrow \{1,\ldots,C\}$ based on these validated scores.k.

C. Enhanced deep convolutional network

As shown in Fig. 2,the network input is an 11-channel tensor $X \in R^{11\times H\times W}$, containing 7 Landsat bands and 4 aligned FCM abundance channels[6]. The network employs a multi-scale feature extraction strategy, first extracting cross-band correlations through cascaded convolutional layers:

$$F_1 = \mathrm{ReLU}(\mathrm{GN}(\mathrm{Conv7} \times 7(X))) \qquad (5)$$

$$F_2 = \mathrm{ReLU}(\mathrm{GN}(\mathrm{Conv5} \times 5(F_1))) \qquad (6)$$

Here, GN represents the group normalization layer, which addresses the problem of unstable training with small batch size.

Subsequently, deep features are extracted through four residual blocks, each designed as follows:

$$\mathrm{ResBlock}(x) = \mathrm{ReLU}(+\mathrm{GN}(\mathrm{Conv3} \times 3(\mathrm{Dropout}(\mathrm{ReLU}(\mathrm{GN}(\mathrm{Conv3} \times 3(x))))))) \qquad (7)$$

The features extracted by the endmember branch ($E \in R^{96\times H\times W}$) are concatenated with the fused features ($F_f \in R^{384\times H\times W}$) at the channel dimension, and then processed through the abundance prediction branch to generate an unnormalized response: Z = Conv_(1×1)(Cat[F_f, E]). The final abundance map A is obtained through pixel-level Softmax normalization:

$$\hat{a}k, h, w = \frac{\exp\left(Z_{k,h,w}\right)}{\sum_{c=1}^{C} \exp\left(Z_{c,h,w}\right)} \qquad (8)$$

Furthermore, the UO-CNN adopts an enhanced composite loss function. This comprehensive loss function ensures that multiple objectives can be optimized simultaneously:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{MSE} + \lambda_2 \mathcal{L}_{MAE} + \lambda_3 \mathcal{L}_{corr} + \lambda_4 \mathcal{L}_{STO} + \lambda_5 \mathcal{L}_{smooth} + \lambda_6 \mathcal{L}_{sparse} \qquad (9)$$

$\mathcal{L}_{corr} = -\frac{1}{N}\sum_i \rho(a_i, a_i)$ is the negative correlation coefficient loss; $\mathcal{L}_{smooth}$ is the spatial smoothness

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constraint, $\mathcal{L}_{STO} = \frac{1}{N}\sum h,w|\sum_k \hat{a}_{k,h,w} - 1|$ is the sum-to-one constraint; $\mathcal{L}_{smooth}$ is the spatial smoothness constraint.

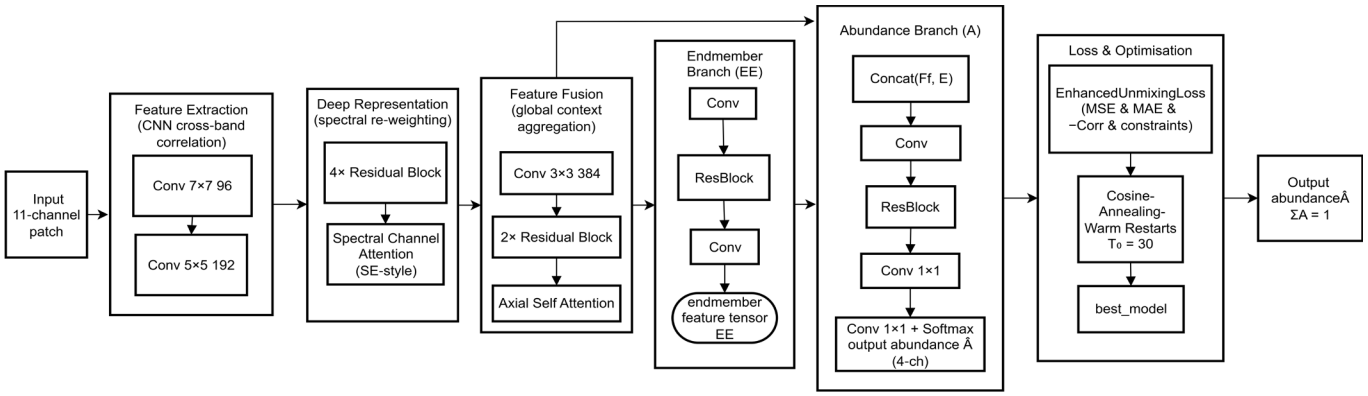


Fig.2. Enhanced dual-input deep convolutional network architecture diagram.

III. Experiment

To comprehensively evaluate the effectiveness of the proposed UO-CNN method, experiments were conducted on six representative research sites. The 30m multispectral Landsat images used as input for SPM were all acquired from the United States Geological Survey. For quantitative evaluation, the fine spatial resolution land cover map of the study areas was generated by classifying the temporally closest 10m Sentinel-2 multispectral images using a Support Vector Machine (SVM).

The method is implemented using Python 3.13.5 and PyTorch 2.7.0, with training conducted on an Intel i7-13700H processor and NVIDIA GeForce GTX 4070 GPU.

A. UO-FCM de-mixing experiment

The experiment validated three representative sites exhibiting significant differences[1]. As shown in Fig. 3, the Unsupervised Optimization method demonstrates notable advantages in preserving terrain boundaries and spatial continuity. Quantitative evaluation employed correlation coefficient, root mean square error, and mean absolute error as metrics. Results indicate that compared to the traditional Linear Spectral Mixture Model, the UO method significantly improved unmixing performance across all test sites: overall regions showed average gains of 2.05% in correlation coefficient with 5.11% RMSE and 4.15% MAE reductions. Critically, mixed-pixel-concentrated areas exhibited even greater improvements: 2.82% higher correlation coefficient, 9.69% lower RMSE, and 8.01% reduced MAE

TABLE I. Estimation performance of LSMM abundance(Overall region).

SITE	R (Raw)	R (New)	RMSE (Raw)	RMSE (New)	MAE (Raw)	MAE (New)
Site1	0.841	0.852	0.205	0.198	0.118	0.115
Site2	0.815	0.836	0.219	0.204	0.126	0.119
Site3	0.790	0.808	0.256	0.243	0.115	0.110

These results demonstrate that the UO method enhances spectral unmixing accuracy through object-based analysis, particularly for complex mixed pixels where it significantly outperforms traditional approaches. This method effectively alleviates the precision limitations of traditional LSMM for complex mixed pixels, providing higher-quality abundance information for subsequent subpixel mapping.

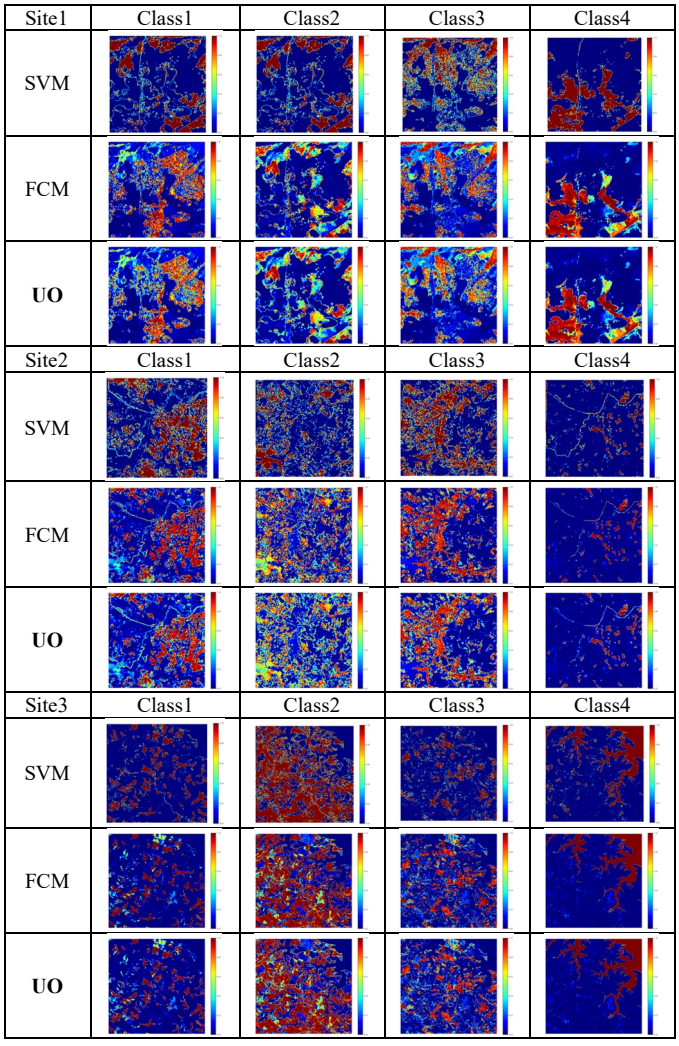


Fig.3. Subpixel unmixing results of three algorithms across three study sites.

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Tables 1 and 2 present the performance comparison results for Overall regions and Mixed regions, respectively.

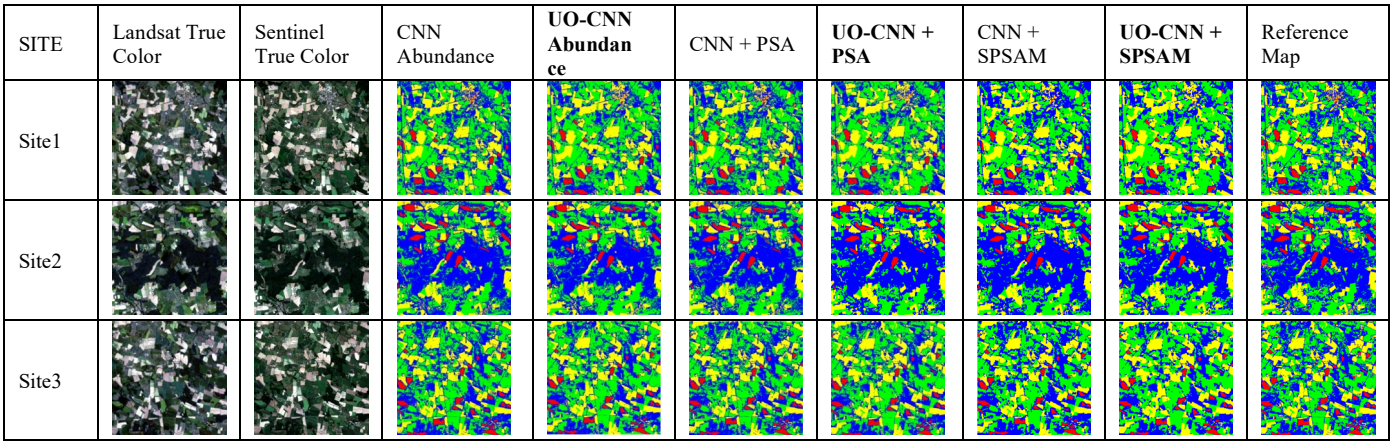


Fig.4. Comparative unmixing and mapping results of CNN and UO-CNN across three study sites.

TABLE II. Performance of LSMM abundance estimation (mixed regions).

SITE	R (Raw)	R (New)	RMSE (Raw)	RMSE (New)	MAE (Raw)	MAE (New)
Site1	0.748	0.758	0.217	0.202	0.136	0.129
Site2	0.655	0.677	0.256	0.228	0.168	0.153
Site3	0.689	0.715	0.339	0.301	0.221	0.199

B. Comparison between UO-CNN and traditional methods

This experiment compared the performance of the UO-CNN and traditional methods in spectral unmixing and subpixel mapping. The experiment was conducted across 3 test regions using a 3-fold scale factor for subpixel mapping, combining PSA and SPSAM mapping algorithms for comprehensive evaluation.

Based on Fig. 4 analysis, the pure CNN abundance maps exhibit obvious blocky artifacts and discontinuities in Sites 1 and 2, while the UO-CNN presents smoother spatial distributions; at land cover boundaries, the pure CNN produces blurred transition areas, whereas the UO-CNN more accurately maintains boundary clarity. For complex mixed pixel regions, the pure CNN tends to yield unstable abundance allocation, while the UO-CNN achieves more reasonable abundance estimation through FCM prior information. Crucially, the UO-CNN+SPSAM yielded results closest to reference maps across all regions, with the UO-CNN outcomes showing superior accuracy in spatial distributions, particularly for the boundary delineation of the new Class 2 (green) and Class 3 (blue).

TABLE III. Comparison of spectral deheterogeneity.

SITE	METHOD	R	RMSE	MAE
Site 1	CNN	0.8937	0.0587	0.0392
	UO-CNN	**0.9241**	**0.0440**	**0.0294**
Site 2	CNN	0.8889	0.0595	0.0397
	UO-CNN	**0.9186**	**0.0446**	**0.0298**
Site 3	CNN	0.8761	0.0671	0.0448
	UO-CNN	**0.9063**	**0.0503**	**0.0336**

As shown in Table 3, quantitative evaluation based on correlation coefficient, root mean square error, and mean absolute error demonstrates that UO-CNN performs excellently across all test areas. Crucially, it exhibits superior generalization capability, maintaining high accuracy even in scenarios containing unknown land cover categories. This robustness is especially evident in the precise identification and boundary mapping of the novel Classes 2 and 3.

IV. Conclusion

This paper proposes an Unsupervised Optimization-Enhanced Convolutional Neural Network (UO-CNN) approach for spectral unmixing. This innovative method synergistically integrates the unsupervised generalization capability of Fuzzy C-Means (FCM) clustering with the fine-grained modeling capacity of deep convolutional neural networks, achieving a harmonious balance between high precision and robust generalization performance.

Future work will explore weaker supervised or unsupervised frameworks to further reduce dependence on high-resolution reference data.

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